Texas A&M University
Department of Electrical and Computer Engineering

ECEN 474/704 – (Analog) VLSI Circuit Design

Fall 2016

Exam #1

Instructor: Sam Palermo

- Please write your name in the space provided below
- Please verify that there are **6 pages** in your exam
- You may use one double-sided page of notes and equations for the exam
- Good Luck!

Problem	Score	Max Score
1		40
2		40
3		20
Total		**100**

Name: _____

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Problem 1 (40 points)

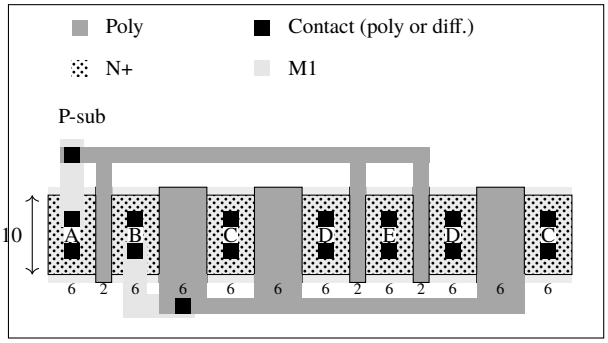
For the layout below, assume that all the commonly labeled diffusion areas are connected with the appropriate metal layers. Assume that $V_A = 2\,\text{V}$, $V_B = 1.2\,\text{V}$, $V_C = 0\,\text{V}$, $V_D = 1.2\,\text{V}$, $V_E = 2\,\text{V}$, $V_{T0} = 0.7\,\text{V}$, $\gamma = 0$, and that all Spice parameters are given (i.e. C_j, C_{jsw}, C_{jc}, C_{ox}, C_{ov}). The dimensions are given in μm, with the poly gates having an $L_D = 0.1\,\mu$m.

a) Draw the equivalent circuit. Combine all parallel transistors and give the total width and length of the equivalent transistors.

b) What region(s) are the transistors operating in?

c) **For node B only**, give an expression and calculate the total gate cap.

d) **For node D only**, give an expression and calculate the total junction cap, including the relevant channel–bulk cap. Note for the perimeter terms, include the sides underneath the gate.



(a) Parallel combinations. The six consecutive devices along the strip are

terminals	A:B	B:C	C:D	D:E	E:D	D:C
W/L	10/2	10/6	10/6	10/2	10/2	10/6

Combining the parallel C–D and D–E devices gives the circuit at right. All W/L values are drawn dimensions in μm/μm.

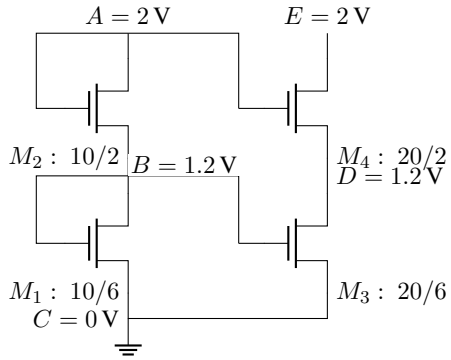
(b) Operating regions. For M_1,

$$V_{GS} = V_{DS} = 1.2\,\text{V} > V_T, \quad V_{DS} > V_{GS} - V_T.$$

Hence M_1 is in saturation. M_3 has the same voltages and is also in saturation. For M_2,

$$V_{GS} = V_{DS} = 0.8\,\text{V} > V_T, \quad V_{DS} > V_{GS} - V_T.$$

Thus M_2 and, similarly, M_4 are in saturation.



(c) Gate capacitance at B. For M_1, its gate–drain overlap is shorted out because both terminals are at B. With $L_{\text{eff}} = 6 - 2L_D = 5.8\,\mu$m,

$$\begin{aligned} C_{gB,M_1} &= \frac{2}{3}(10)(5.8)\,\mu\text{m}^2 C_{ox} + 10\,\mu\text{m} C_{ov} \\ &\simeq 38.7\,\mu\text{m}^2 C_{ox} + 10\,\mu\text{m} C_{ov}. \end{aligned}$$

For M_3, both overlap capacitors contribute:

$$\begin{aligned} C_{gB,M_3} &= \frac{2}{3}(20)(5.8)\,\mu\text{m}^2 C_{ox} + 2(20)\,\mu\text{m} C_{ov} \\ &\simeq 77.3\,\mu\text{m}^2 C_{ox} + 40\,\mu\text{m} C_{ov}. \end{aligned}$$

$$\boxed{C_{gB} = 116\,\mu\text{m}^2 C_{ox} + 50\,\mu\text{m} C_{ov}.}$$

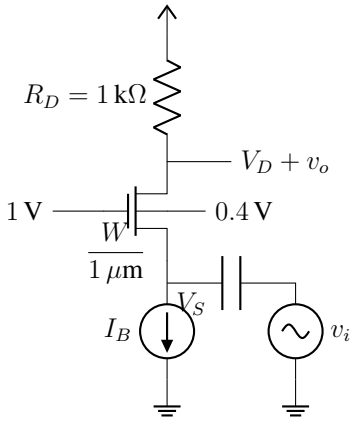
(d) Junction capacitance at D. The two D diffusion regions each have area $6\times 10\,\mu\text{m}^2$ and perimeter $2(6+10)\,\mu$m, including the gate-adjacent edges. Add the channel–bulk term of M_4, with $L_{\text{eff}} = 2 - 2L_D = 1.8\,\mu$m:

$$\begin{aligned} C_{jD} &= A_D C_j + P_D C_{jsw} + C_{cb,M_4} \\ &= (6)(10)(2)\,\mu\text{m}^2 C_j + (6+10)(2)(2)\,\mu\text{m} C_{jsw} + \frac{2}{3}(20)(1.8)\,\mu\text{m}^2 C_{jc} \\ &= \boxed{120\,\mu\text{m}^2 C_j + 64\,\mu\text{m} C_{jsw} + 24\,\mu\text{m}^2 C_{jc}.} \end{aligned}$$

Problem 2 (40 points)

For the circuit shown below, the transistor is operating in a fully velocity saturated region where its DC drain current is given exactly by the given equation.

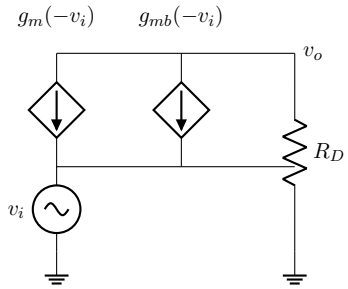
a. Draw the low-frequency small-signal model of the circuit (neglecting transistor capacitors) and give an expression for the small-signal voltage gain of the circuit, $A_v = v_o/v_i$, as a function of the relevant transistor conductances and the load resistor. Note, you can assume that the input AC-coupling capacitor acts as an ideal short.



$$I_{DS} = v_{sat}C_{ox}W(V_{GS} - V_{Tn})$$

$$V_{Tn} = V_{T0} + \gamma\left(\sqrt{|2\Phi_F| + V_{SB}} - \sqrt{|2\Phi_F|}\right).$$

Typeset solution. There is no V_{DS} dependence in the stated model, so $g_{ds} = 0$ and $r_o = \infty$.



$$g_m = \frac{\partial I_D}{\partial V_{GS}} = v_{sat}C_{ox}W,$$

$$g_{mb} = \frac{\partial I_D}{\partial V_{BS}} = v_{sat}C_{ox}W(-1)(-1)\frac{\gamma}{2\sqrt{|2\Phi_F| + V_{SB}}} = \eta g_m,$$

$$\eta = \frac{\gamma}{2\sqrt{|2\Phi_F| + V_{SB}}}.$$

$$v_o = -(g_m + g_{mb})(-v_i)R_D, \qquad \boxed{A_v = \frac{v_o}{v_i} = (g_m + g_{mb})R_D.}$$

b. Assume the following DC operating points for the drain and source voltages, $V_{S,Q} = 0.2\,\text{V}$ and $V_{D,Q} = 1\,\text{V}$, and that $v_{sat}C_{ox} = 250\,\mu\text{A}/(\text{V}\cdot\mu\text{m})$, $V_{T0} = 0.7\,\text{V}$, $\gamma = 0.45\,\text{V}^{1/2}$, $2\Phi_F = 0.9\,\text{V}$, and $L_D = 0.1\,\mu\text{m}$. Calculate the necessary transistor width W for $|A_v| = 5\,\text{V}/\text{V}$.

Typeset solution. Here $V_{SB} = 0.2 - 0.4 = -0.2\,\text{V}$. Consequently,

$$\eta = \frac{0.45}{2\sqrt{0.9 - 0.2}} \simeq 0.269.$$

$$A_v = (1+\eta)g_m R_D = (1+\eta)v_{sat}C_{ox}WR_D,$$

$$\boxed{W = \frac{A_v}{(1+\eta)v_{sat}C_{ox}R_D} = \frac{5}{(1.269)(250\,\mu\text{A}/(\text{V}\cdot\mu\text{m}))(1\,\text{k}\Omega)} \simeq 15.8\,\mu\text{m}.}$$

Problem 3 (20 points)

a. Poly resistor design

Design and sketch a layout of a $500\,\Omega$ poly resistor. Assume that the poly $R_\square = 60\,\Omega/\square$, each $(1\,\mu\text{m} \times 1\,\mu\text{m})$ contact has resistance of $10\,\Omega$, and fabrication tolerances limit the minimum unit resistor width $W \geq 3\,\mu$m. Use at least 2 fingers in the resistor design. In the sketch clearly label the critical dimensions.

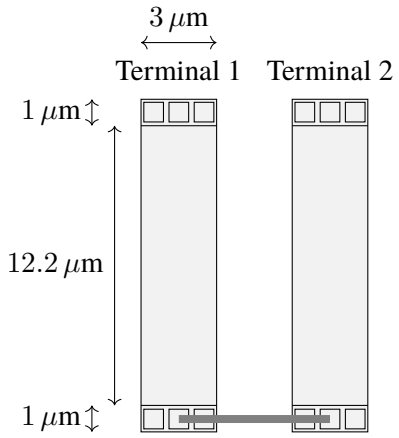
Typeset solution.

$$R_{\text{total}} = \left(\frac{2R_c}{n} + R_\square \frac{L}{W}\right) F, \qquad \begin{array}{l} n = \text{number of parallel contacts per end,} \\ F = \text{number of fingers (unit resistors).} \end{array}$$

$$L = \left(\frac{R_{\text{total}}}{F} - \frac{2R_c}{n}\right) \frac{W}{R_\square}.$$

Set $W = 3\,\mu$m, $n = 3$, and $F = 2$. Then

$$L = \left(\frac{500}{2} - \frac{2(10)}{3}\right)\left(\frac{3\,\mu\text{m}}{60}\right) \simeq \boxed{12.2\,\mu\text{m}}.$$



The source rounds the calculated body length $12.1667\,\mu$m to $12.2\,\mu$m; the capacitance below uses that rounded drawn dimension.

b. Poly resistor parasitic capacitance

The poly resistor has a parallel-plate parasitic capacitance to substrate of $0.1\,\text{fF}/\mu\text{m}^2$. What is the total parasitic capacitance of the resistor? Include the contact area in the capacitance calculation and assume that the capacitance is only due to the parallel-plate effect, i.e. neglect any fringing capacitance.

Typeset solution.

$$\text{Resistor area} = (12.2\,\mu\text{m} + 2\,\mu\text{m})(3\,\mu\text{m})(2) = 86.4\,\mu\text{m}^2.$$

$$\boxed{C_{\text{parasitic}} = (86.4\,\mu\text{m}^2)(0.1\,\text{fF}/\mu\text{m}^2) = 8.64\,\text{fF}.}$$

c. N-well resistor design

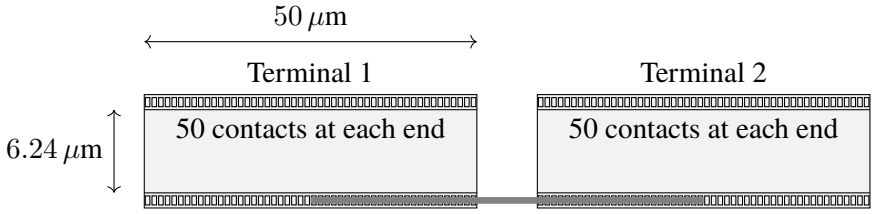
Design and sketch a layout of a $500\,\Omega$ n-well resistor. Assume that the n-well $R_\square = 2\,\text{k}\Omega/\square$, each $(1\,\mu\text{m} \times 1\,\mu\text{m})$ contact has resistance of $10\,\Omega$, and fabrication tolerances limit the minimum unit resistor length $L \geq 6\,\mu$m. Use at least 2 fingers in the resistor design. In the sketch clearly label the critical dimensions.

Typeset solution.

$$R_{\text{total}} = \left(\frac{2R_c}{n} + R_\square \frac{L}{W}\right) F, \qquad L = \left(\frac{R_{\text{total}}}{F} - \frac{2R_c}{n}\right) \frac{W}{R_\square}.$$

Set $F = 2$, $W = 50\,\mu$m, and $n = 50$. Thus

$$L = \left(\frac{500}{2} - \frac{2(10)}{50}\right)\left(\frac{50\,\mu\text{m}}{2000}\right) = \boxed{6.24\,\mu\text{m}}.$$



d. N-well resistor parasitic capacitance

The n-well resistor has a parallel-plate parasitic capacitance to substrate of $1\,\text{fF}/\mu\text{m}^2$. What is the total parasitic capacitance of the resistor? Include the contact area in the capacitance calculation and assume that the capacitance is only due to the parallel-plate effect, i.e. neglect any fringing capacitance.

Typeset solution.

$$\text{Resistor area} = (6.24\,\mu\text{m} + 2\,\mu\text{m})(50\,\mu\text{m})(2) = 824\,\mu\text{m}^2.$$

$$\boxed{C_{\text{parasitic}} = (824\,\mu\text{m}^2)(1\,\text{fF}/\mu\text{m}^2) = 824\,\text{fF}.}$$

e. If we have to use the $500\,\Omega$ resistor for a high-frequency application, which design should we choose, the poly or n-well design? Why?

Typeset solution. Choose the **poly resistor**: it has less parasitic capacitance.

Scratch Paper